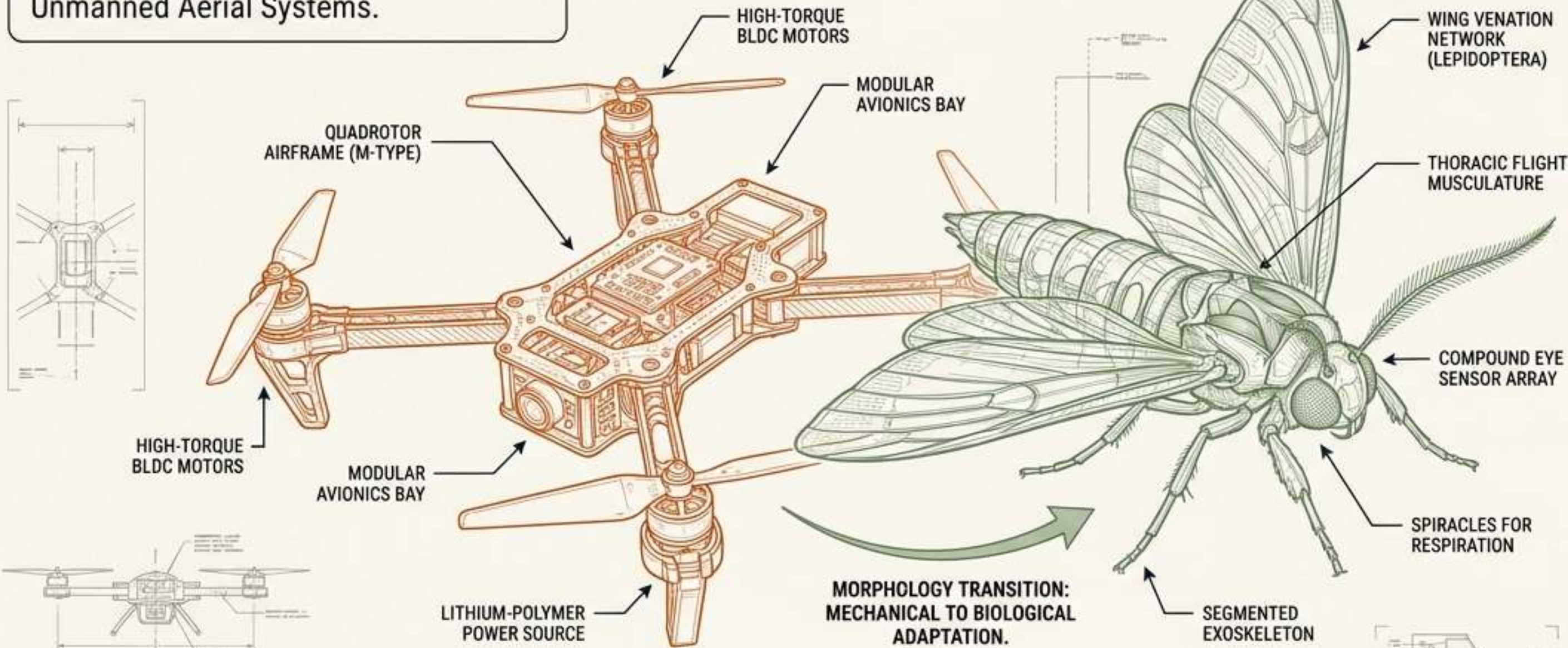


Module 7.2: Bio-Inspired and Next-Gen Drones

Advanced Engineering of Unmanned Aerial Systems.

BIO-INSPIRED QUADROTOR MORPHING SYSTEM: AERIAL ADAPTABILITY.



SYSTEM SPECIFICATIONS (EST.)

WINGSPAN (MECH):	650 mm
WINGSPAN (BIO):	820 mm
MTOW:	1.8 kg
FLIGHT TIME:	~45 min
MAX SPEED:	90 km/h

KEY TECHNOLOGIES

- SHAPE-MEMORY ALLOYS
- DIELECTRIC ELASTOMER ACTUATORS
- NEUROMORPHIC FLIGHT CONTROL
- ADAPTIVE STRUCTURAL COMPOSITES

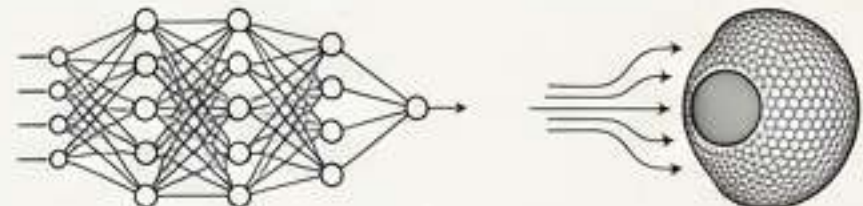
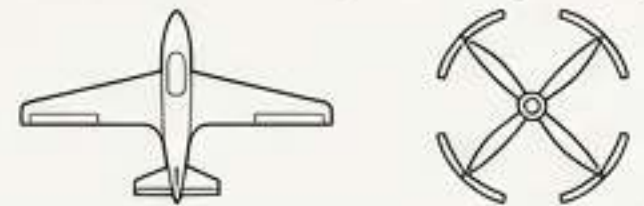
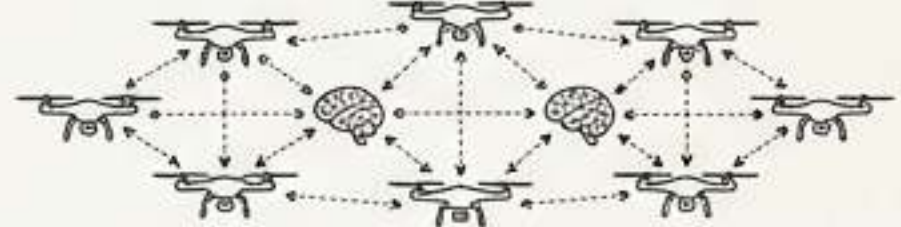
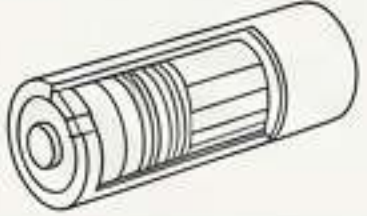
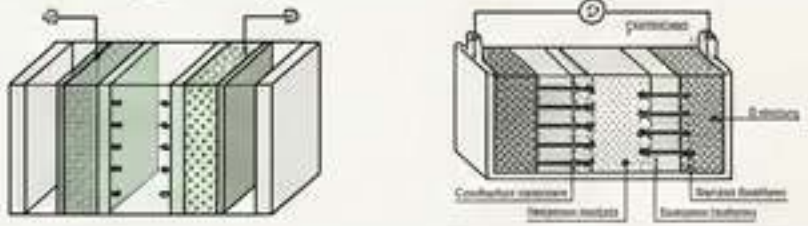
BIO-MIMICRY PRINCIPLES

- PASSIVE WING TWISTING
- ELASTIC ENERGY STORAGE
- HOVERING STABILITY
- DYNAMIC FLIGHT ENVELOPE

MASTERCLASS SERIES // TARGET AUDIENCE: AEROSPACE & AUTONOMOUS SYSTEMS ARCHITECTS.

Integration of Lepidoptera Flight Mechanics for Enhanced Maneuverability and Efficiency in Complex Environments.

COMPARATIVE ANALYSIS: TRADITIONAL VS. BIO-MIMETIC AUTONOMOUS PARADIGMS

	Traditional Rigid-Body 	Bio-Mimetic Autonomous Agents 
Navigation Paradigm	GPS/Frame-based Vision 	Neuromorphic/Optical Flow 
Actuation & Morphology	Static fixed/rotary wings 	4D Morphing structures (Shape Memory Polymers) & Reciprocating Wings (Ornithopters) 
Intelligence Model	Centralized command/Cloud computing 	Decentralized Swarm Intelligence (Edge AI) 
Energy Constraint	Lithium-Ion limits 	Solid-State Hydrogen & Semi-Solid Batteries 

Biology as the ultimate optimization engine: leveraging functional complementarity to overcome **SWaP-C** (Size, Weight, Power, and Cost) limitations.

Biomimetics

Emulating the mechanical, sensory, and collective efficiencies of marine and terrestrial organisms. Fault tolerance and adaptability.

Smart Materials

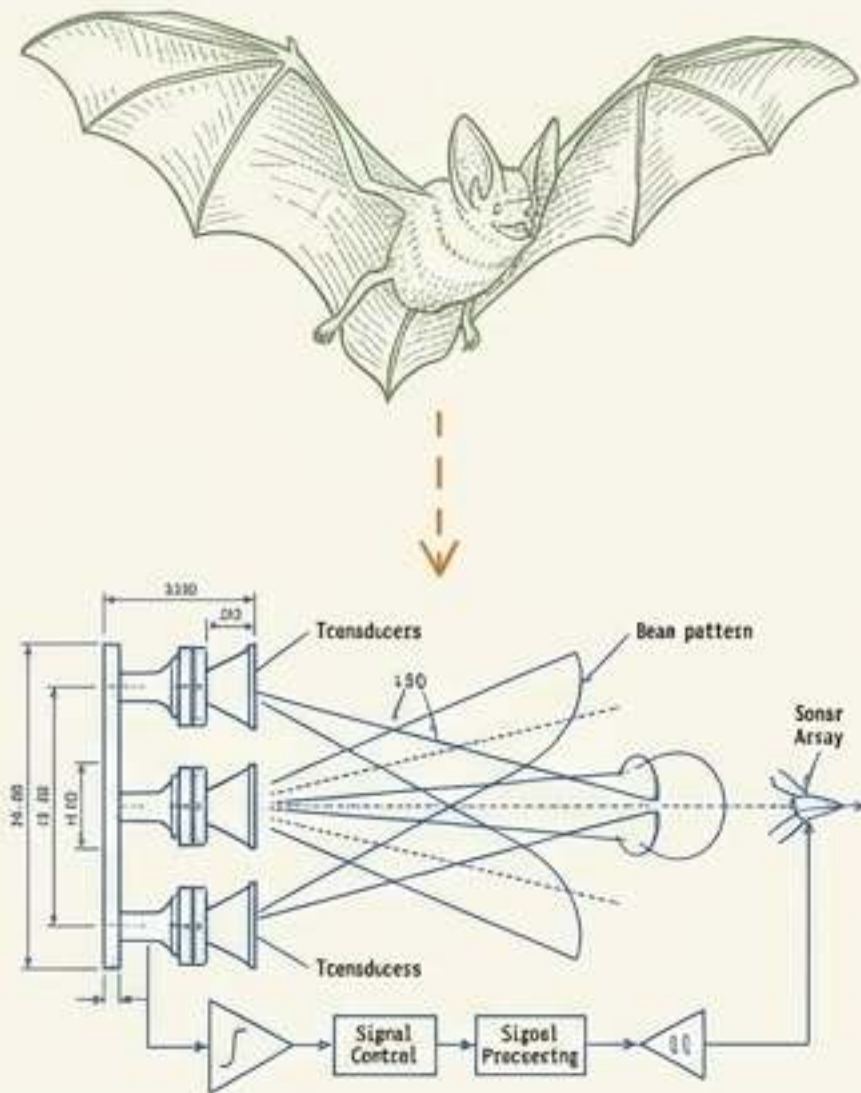
Biodegradable Shape Memory Polymers (SMPs), 4D printing, and variable stiffness structures.

The Cybernetic Agent: A platform capable of transitioning flight modes like an insect, navigating via echolocation, and operating within a self-organizing swarm.

Edge Computing

On-device inference, Spiking Neural Networks (SNNs), and ultra-low-latency processing without cloud reliance.

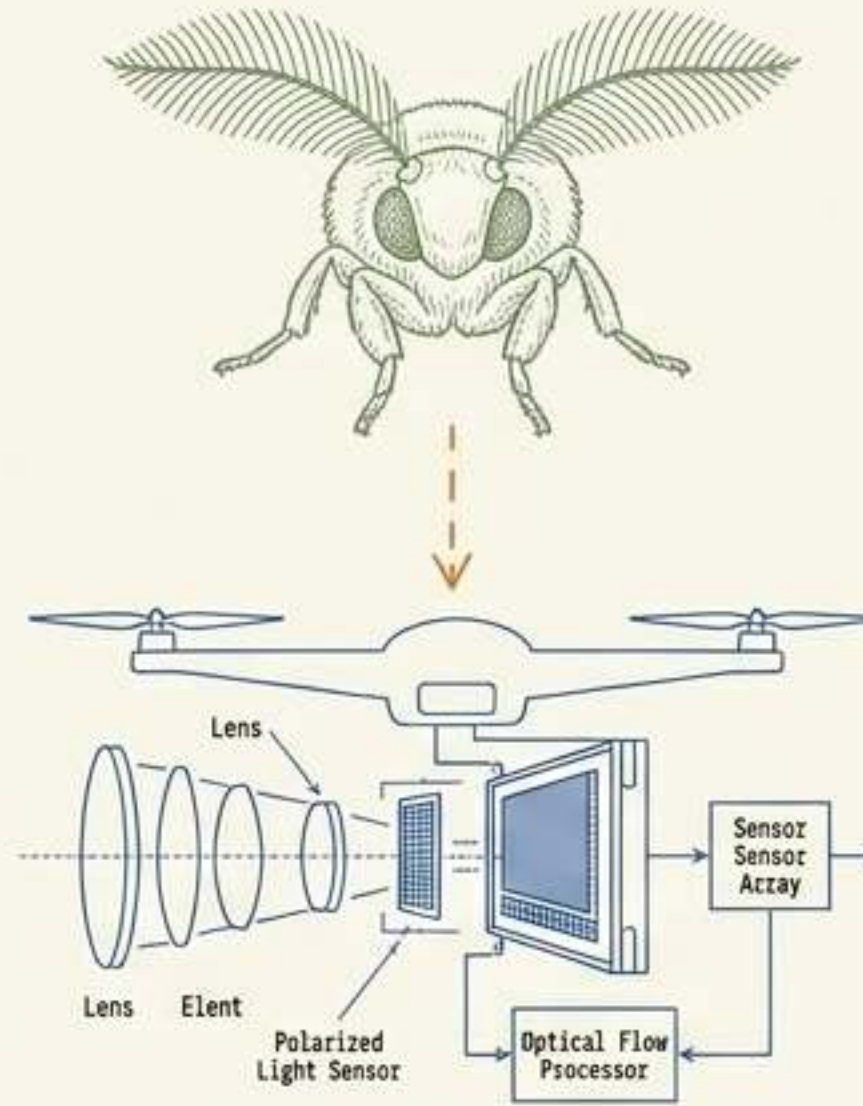
Bats to Active Sonar



Bio-Inspiration: Echolocation for inertial 3D flight in real-time.

Tech Execution: Broadband LPM (Linear Period Modulated) Ultrasonic Pulses. Overcomes negative Doppler shifts and intense propeller noise.

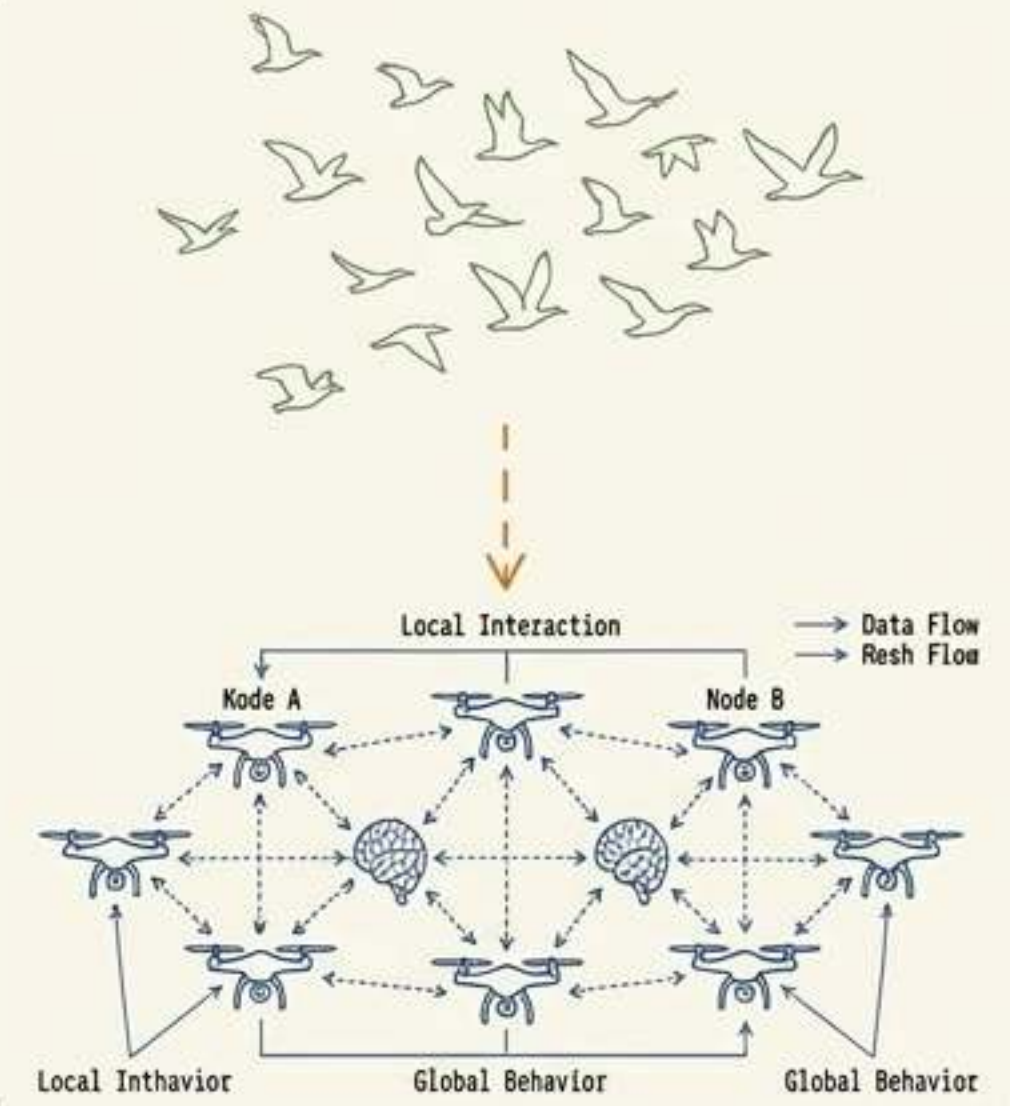
Insects to Vision



Bio-Inspiration: Silkworm moth antennae and desert ant polarized light sensors.

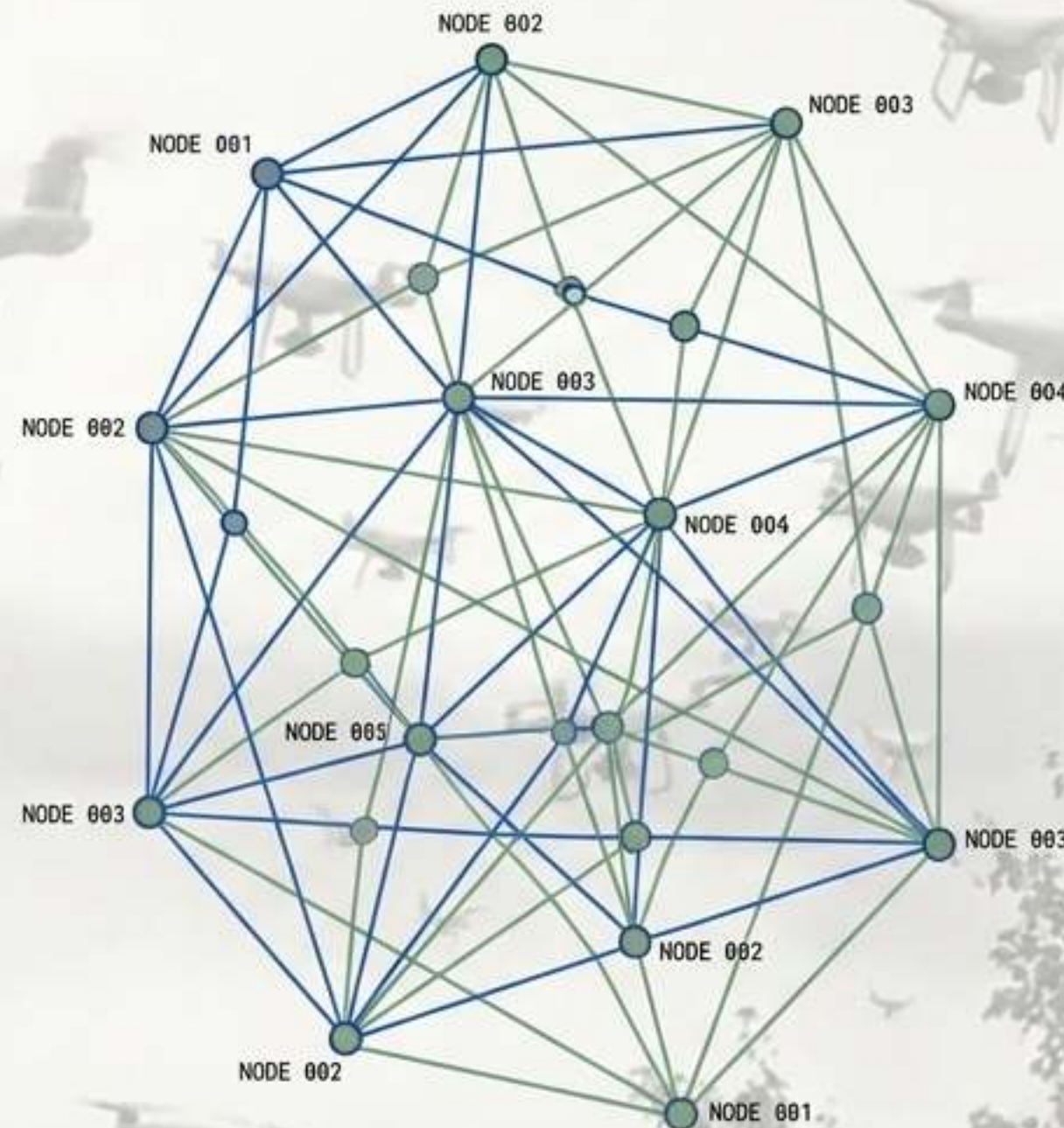
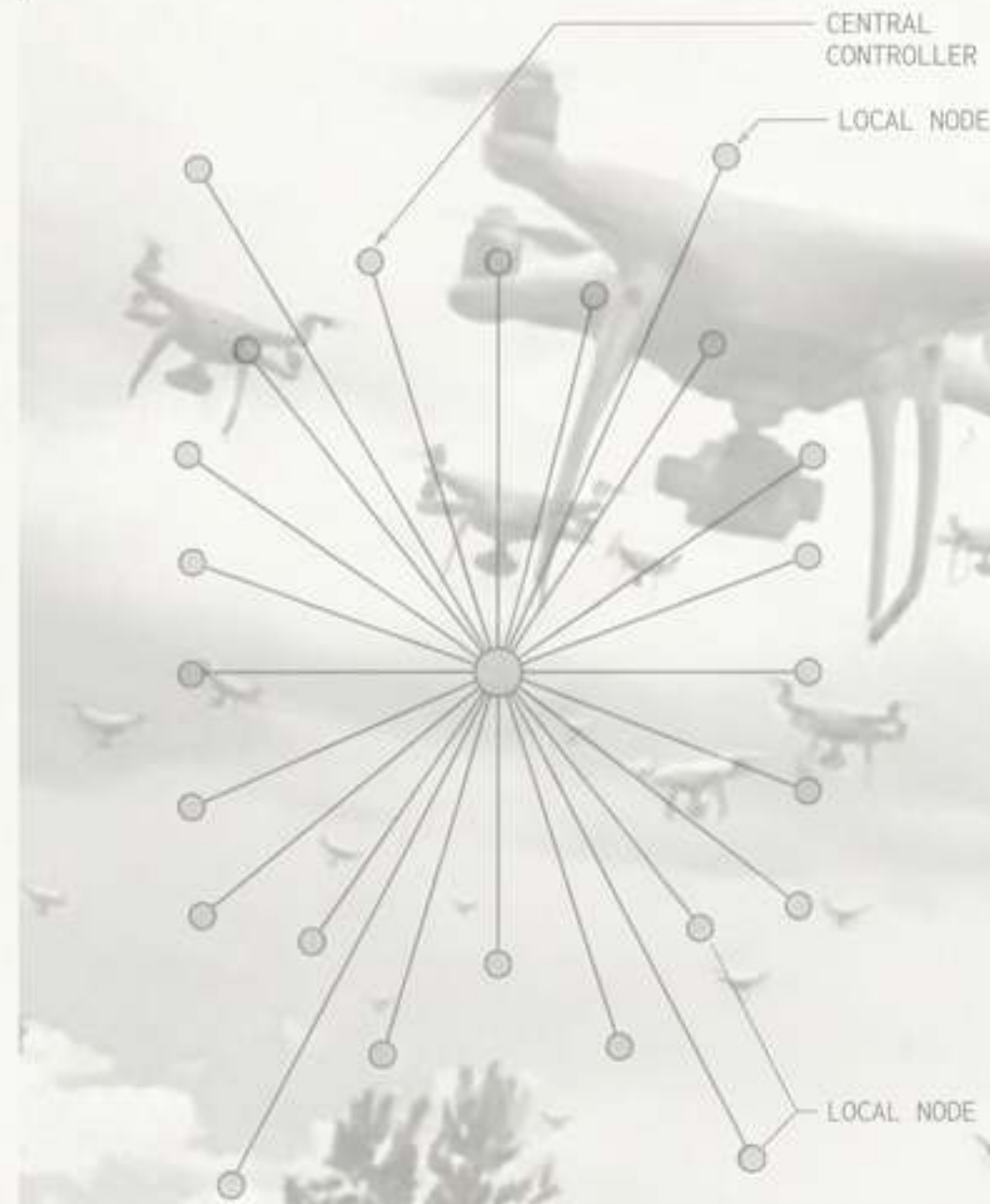
Tech Execution: Optical flow navigation sensors. Low-latency, GNSS-denied spatial awareness.

Flocking to Coordination



Bio-Inspiration: Emergent global behaviors from local interactions.

Tech Execution: Decentralized SWARM algorithms mapped to discrete combinatorial domains.



Particle Swarm Optimization (PSO)

Particles moving through a hyperdimensional search space governed by **cognitive** (individual best) and **social** (swarm best) components.

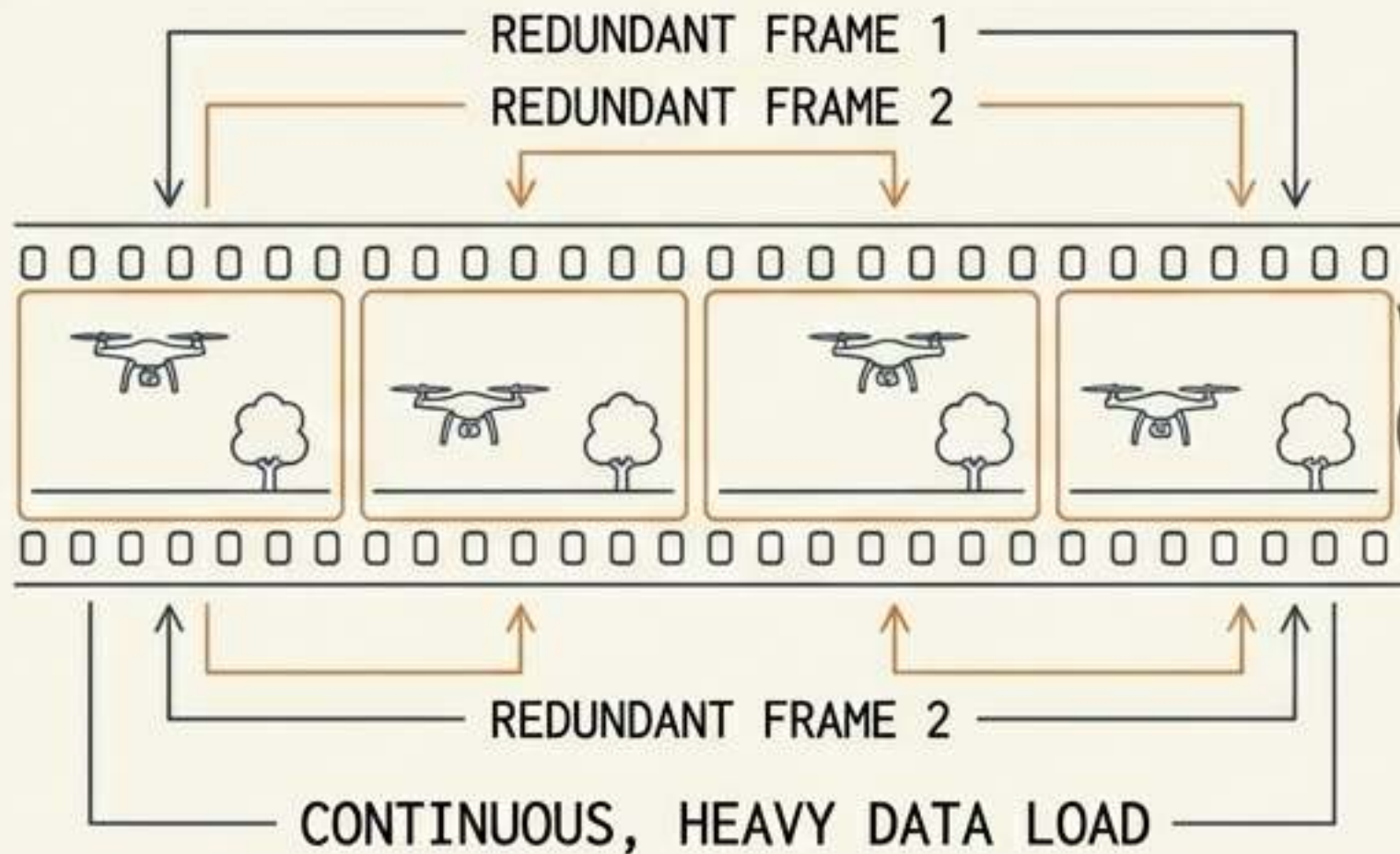
Ant Colony Optimization (ACO)

Leveraging **stigmergy**—indirect coordination through environmental modification (digital pheromones).

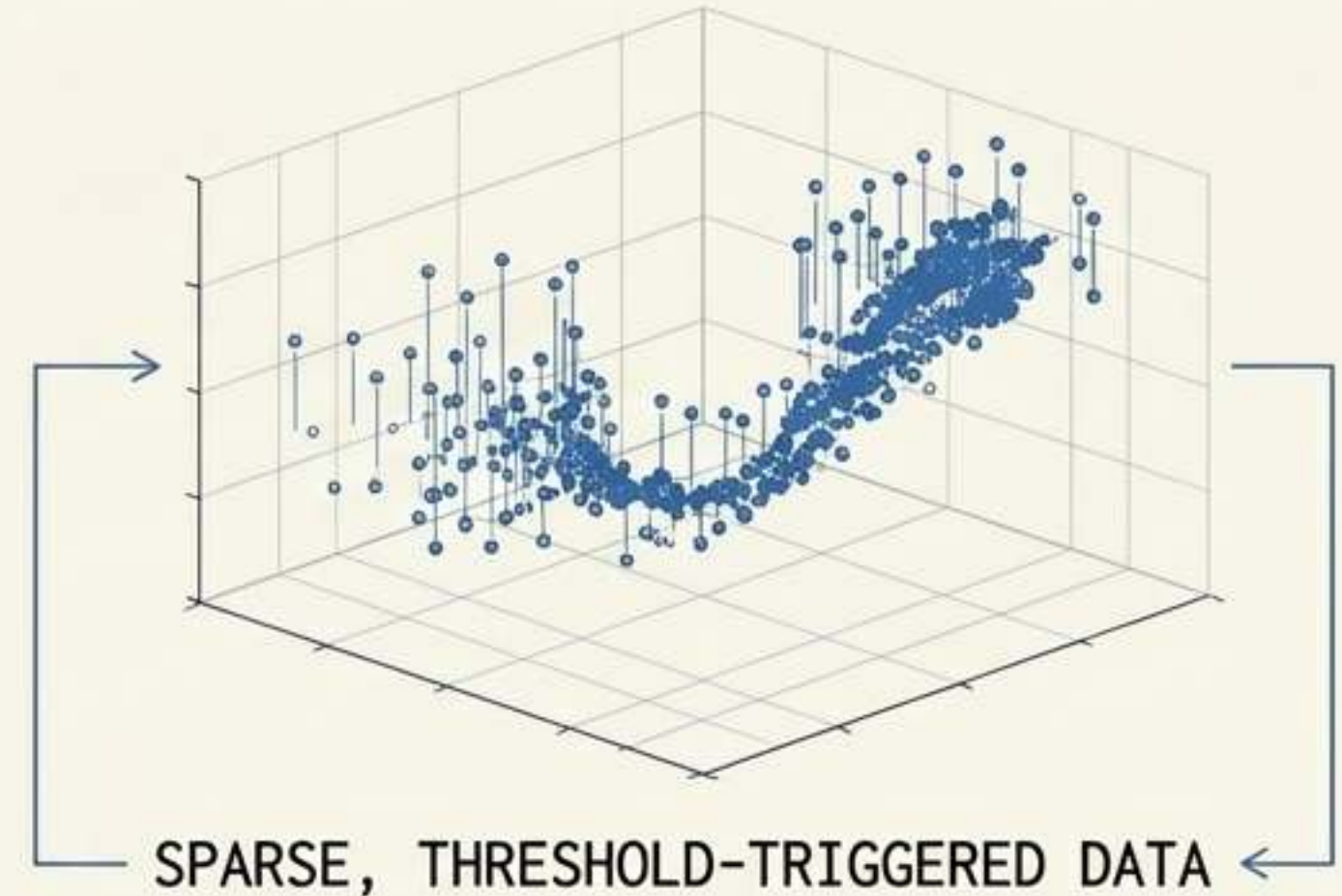
Adapting from continuous optimization problems to discrete domain adaptation for UAV task allocation: assigning specific drones to specific **combinatorial** targets autonomously.

NEUROMORPHIC VISION: FRAME-BASED VS. EVENT-BASED

FRAME-BASED VISION



ASYNCHRONOUS EVENT-BASED PROCESSING



THE MECHANICS:

Neuromorphic/Dynamic Vision Sensors (DVS) report only per-pixel brightness changes asynchronously.

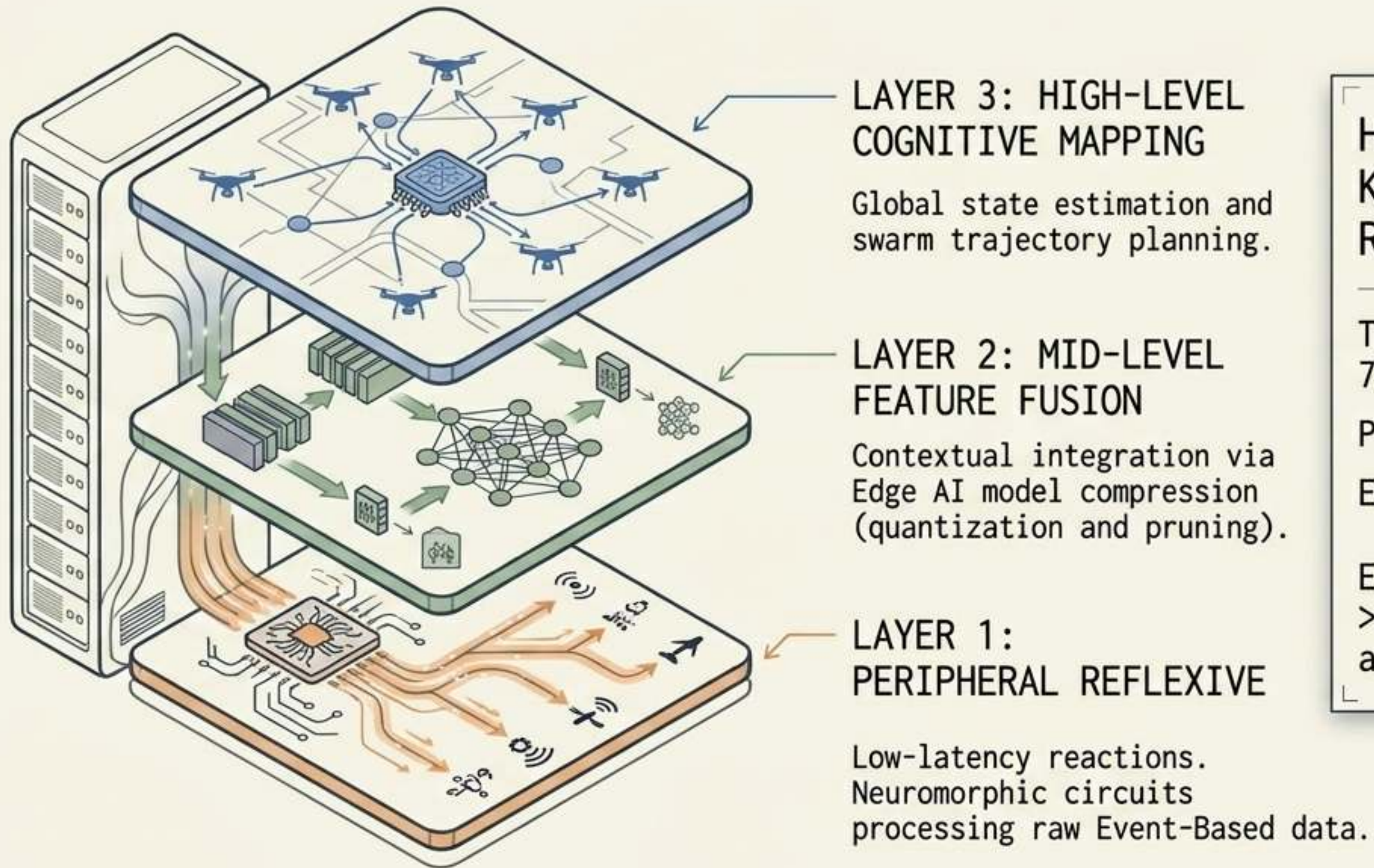
THE DATA PACKET:

Each pixel operates independently, generating a sparse data packet containing:
[Address | Timestamp | Polarity]

THE ADVANTAGE:

Microsecond-level reaction times, high dynamic range (HDR) for extreme lighting changes, and drastically reduced computational payload.

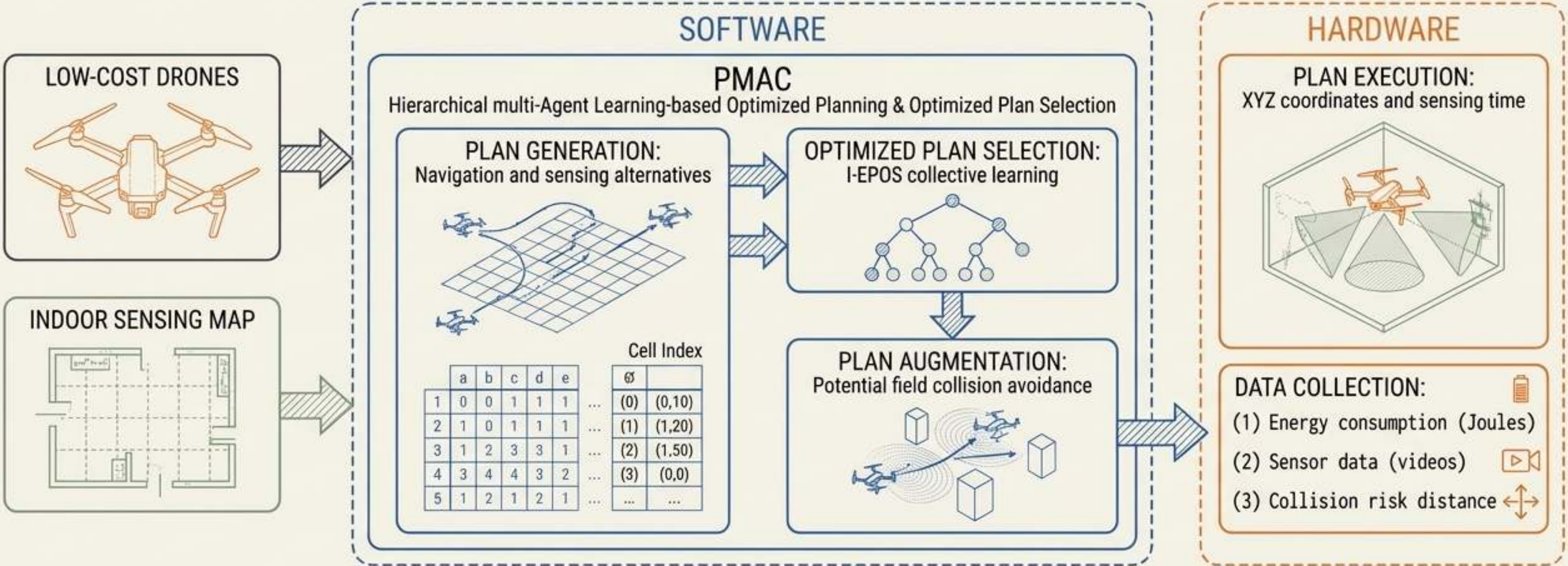
CYBERNETIC NERVOUS SYSTEM STACK & HARDWARE ACCELERATION



HARDWARE SPOTLIGHT:
KRAKEN/COLIBRIUAV
RISC-V SOC

Throughput:
7200 event frames/sec
Power Consumption: 10.7 mW
Efficiency:
Efficiency:
>6.6x faster than RGB
architectures

PROTOTYPED M-TET ARCHITECTURE & OPTIMIZED PLANNING



HALOP (Hierarchical multi-Agent Learning-based Optimized Planning)

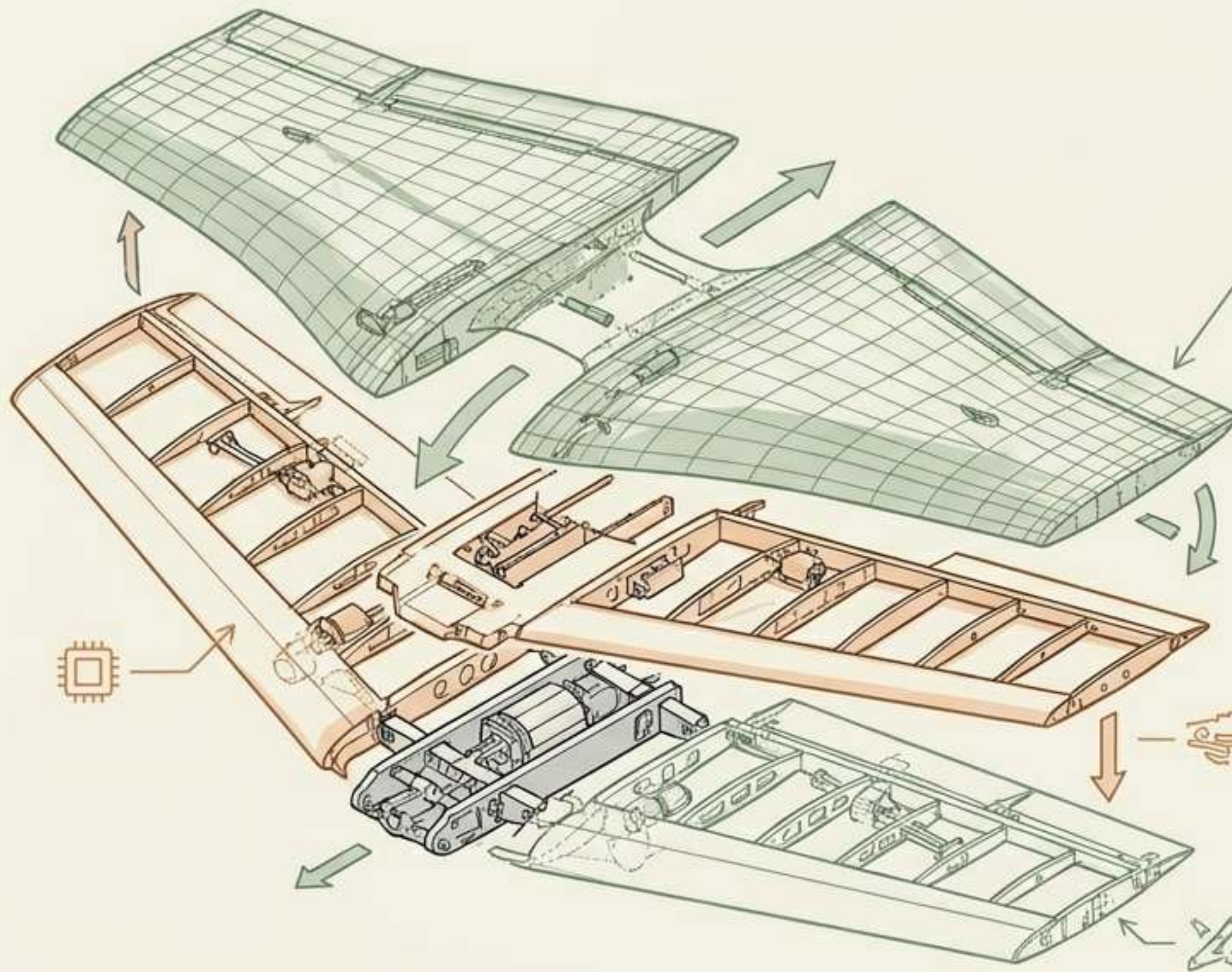
The learning-based engine for high-level strategic decisions, dynamically adapting to constraints.

OPS (Optimized Plan Selection)

The static optimization engine utilizing exact algorithms (greedy) to ensure hard constraints and fine-grained task allocation are met.

KEY TAKEAWAY: Combining deep reinforcement learning with exact combinatorial optimization drastically reduces the decision space and improves scalability.

MORPHING DRONE WING: ADAPTIVE MATERIAL SCIENCE & ORNITHOPTERS



4D PRINTING & SHAPE MEMORY POLYMERS (SMPs)

Biodegradable smart polymers that change shape in response to environmental stimuli (temperature, aerodynamic pressure) without discrete mechanical actuators.

ORNITHOPTERS (RECIPROCATING WINGS)

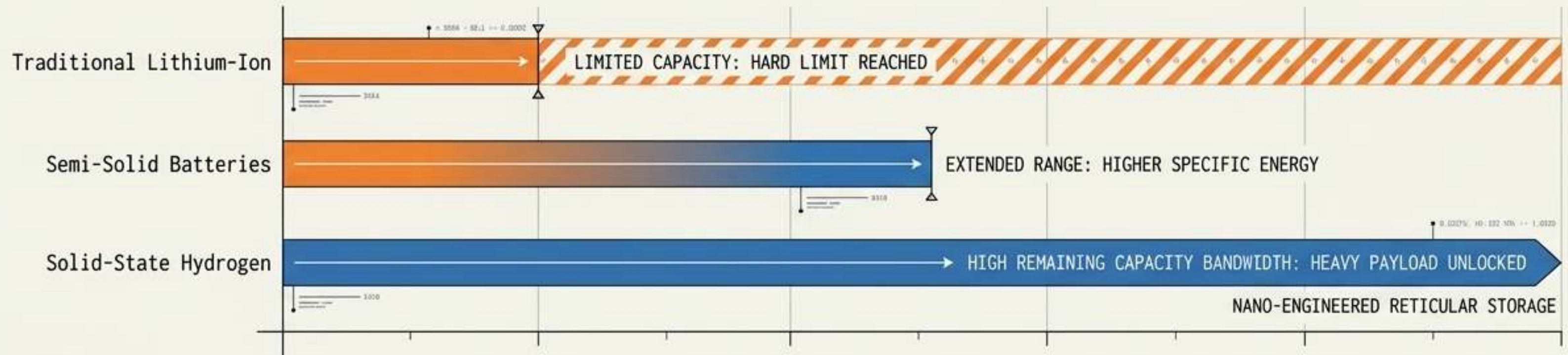
Emulating unsteady aerodynamic modeling and flapping wing lift/thrust efficiency to navigate constrained environments.

OPERATIONAL BENEFIT

Combines the structure and actuator into a single element, minimizing weight, reducing mechanical complexity, and avoiding catastrophic part failure.

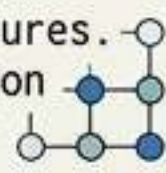
PROPULSION BREAKTHROUGHS & ENERGY DENSITY COMPARISON

SPECIFIC ENERGY DENSITY (Wh/kg) COMPARISON



SOLID-STATE HYDROGEN

Utilizing nano-engineered reticular materials to store hydrogen at low pressure and near-ambient temperatures. Unlocks heavy payload, long-duration missions.



SEMI-SOLID BATTERIES

Bridging the gap between traditional liquid electrolytes and full solid-state, offering higher specific energy and safety.



ENVIRONMENTAL IMPACT

Zero-emission energy aligned with green aviation and sustainable deployment.



COMPLEX TERRAIN SAR: DRONE SWARM COORDINATION & M-TET VALIDATION

Application Scenario: Complex Terrain SAR

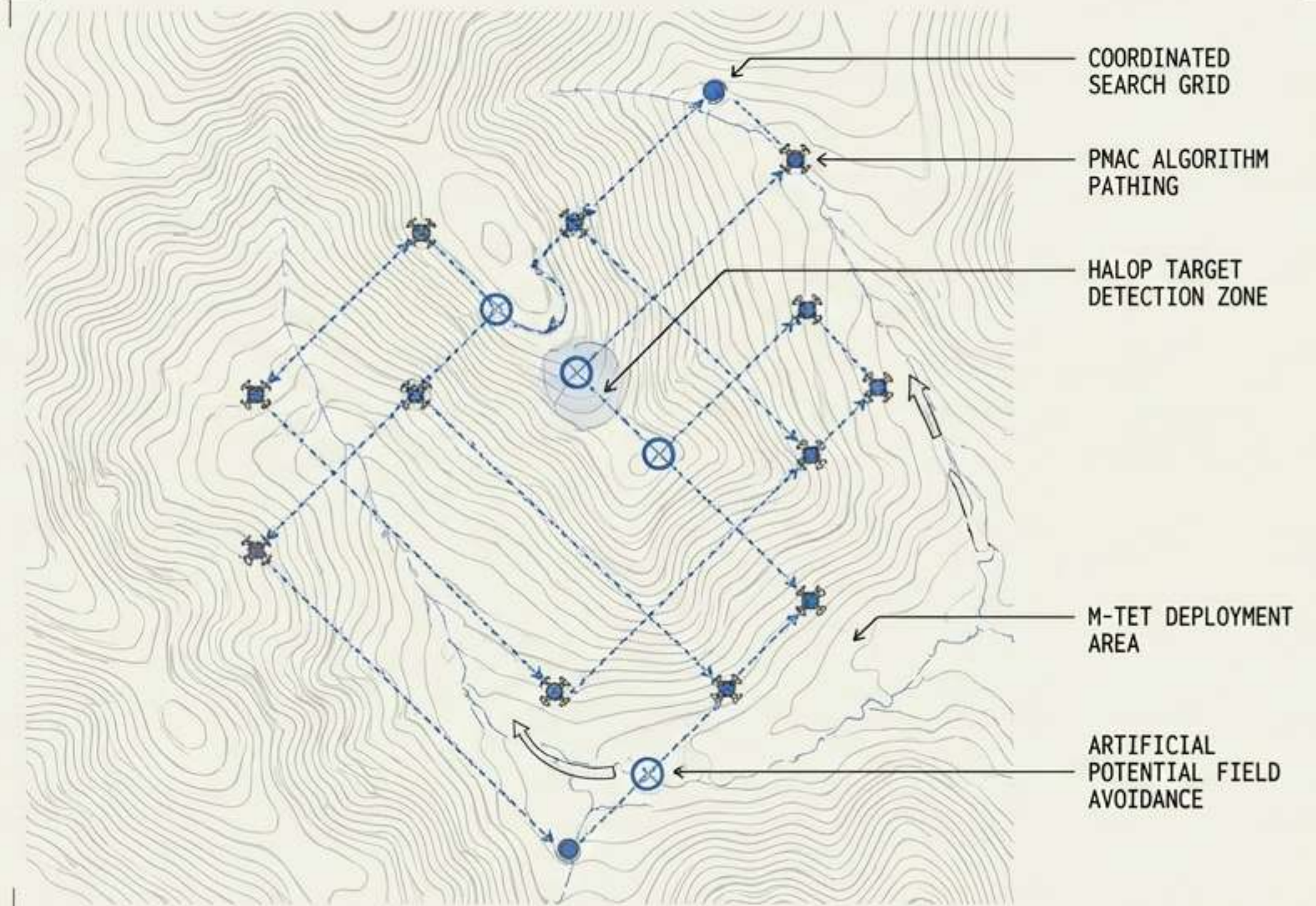
Applying PMAC and HALOP algorithms to rapidly scan, detect, and escort targets in unpredictable disaster environments.

Validation Framework: M-TET

The Multi-drone Tasking Experimentation Testbed. Bridging simulation and physical deployment. Utilizing low-cost hardware to validate energy-aware sensing, trajectory coordination, and Artificial Potential Field collision avoidance.

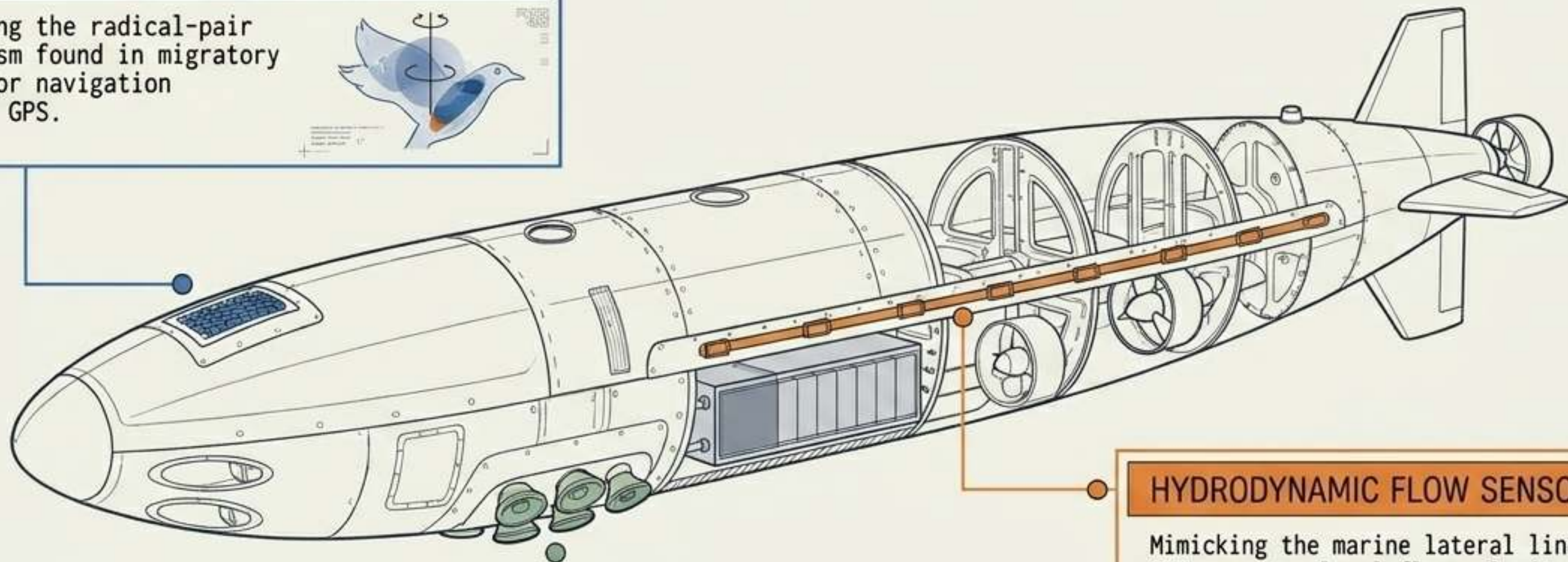


LOW-COST HARDWARE REFERENCE:
M-TET DRONE UNIT



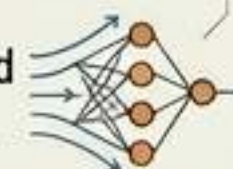
QUANTUM-ASSISTED MAGNETORECEPTION

Emulating the radical-pair mechanism found in migratory birds for navigation without GPS.



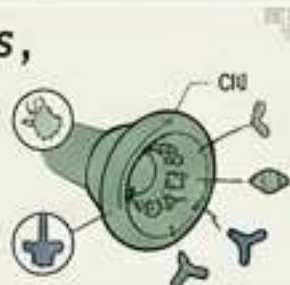
HYDRODYNAMIC FLOW SENSORS

Mimicking the marine lateral line. SNNs process local flow velocity variations, triggering event packets only when changes exceed spatial thresholds.



TACTILE-CHEMICAL SUCKERS

Inspired by *Octopus vulgaris*, providing decentralized somatotopic object identification.



MULTIMODAL SENSOR FUSION ARCHITECTURE FOR UNSTRUCTURED ENVIRONMENTS



SCALE REFERENCE:
STANDARD QUARTER

Miniaturization Limits

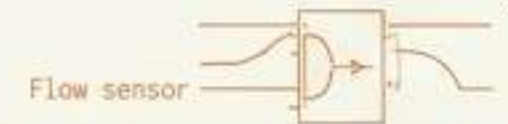
Pushing flight physics toward the biological limits of insects (bees/moths).

Sub-Gram Avionics

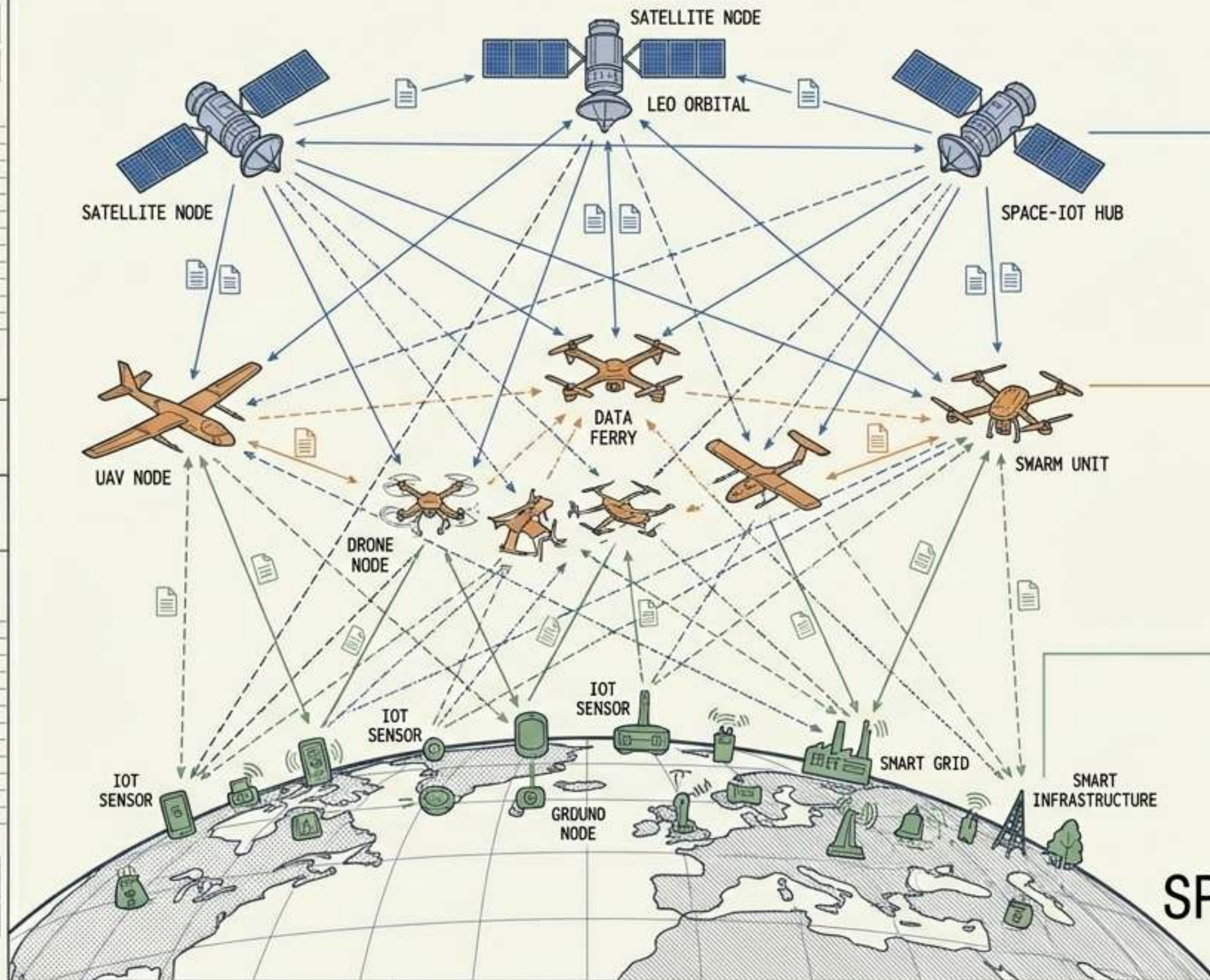


Utilizing sub-gram optical flow sensors and rad-hard CDH (Command and Data Handling) units for onboard processing without heavy high-resolution visual sensors.

Applications



Ubiquitous mobile sensor networks for stealth reconnaissance and microscopic infrastructure inspection.



Internet-of-Space-Things

Moving beyond isolated operations to integrated, continuous over-the-air multi-agent learning.

Edge-to-Cloud Offloading

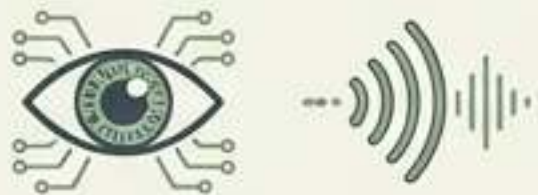
Drones serving as data ferries, executing 'download while charging' loops.

Industry 5.0 Integration

Human-centric industrial systems where autonomous swarms complement human decision-making, utilizing Digital Twins for real-time predictive maintenance.

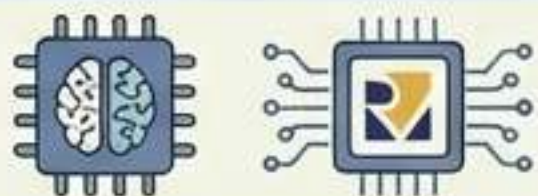
6G-ENABLED INTERNET-OF-SPACE-THINGS ARCHITECTURE

Sensory: Neuromorphic DVS & LPM Echolocation

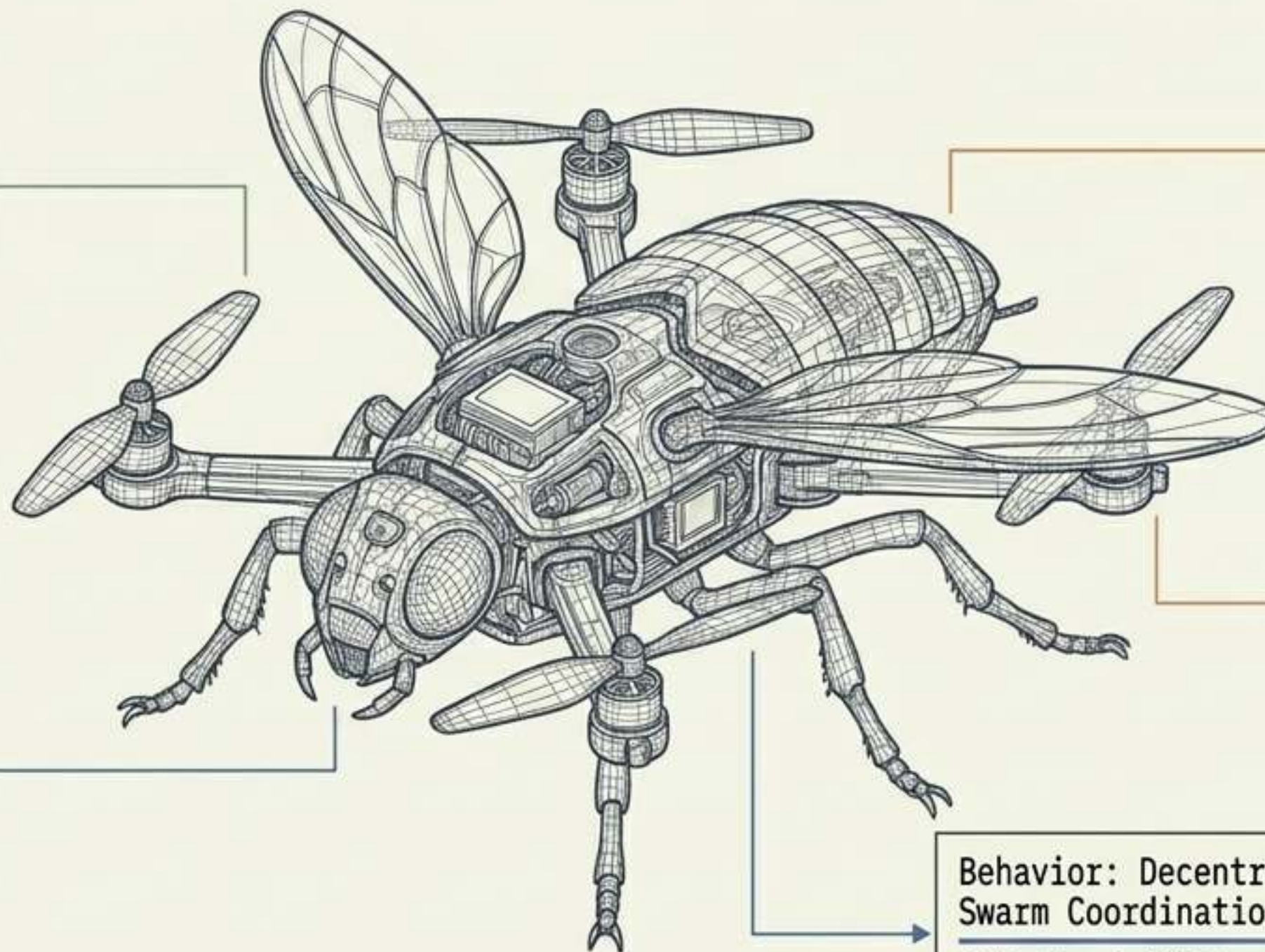


Bio-inspired vision and sonar systems for low-power, high-dynamic range sensing in complex environments.

Nervous System: Kraken RISC-V Edge AI & PMAC Framework



Distributed neuromorphic computing and parallel multi-agent control for real-time edge intelligence.

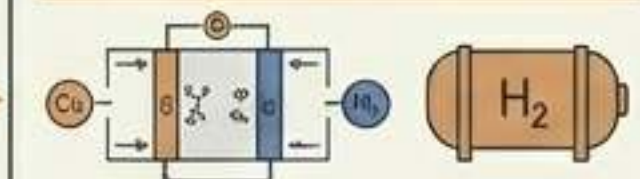


Musculature: 4D SMP Morphing Wings



Active wing morphing capabilities for optimized flight efficiency across varied regimes.

Metabolism: Solid-State Hydrogen Fuel



High energy density solid-state hydrogen storage and power generation for extended endurance.

Behavior: Decentralized PSO/ACO Swarm Coordination



Collective intelligence algorithms enabling robust, scalable, and autonomous swarm operations.

UAVs are no longer mechanical tools; they are evolving into highly sophisticated, bio-mimetic autonomous agents, redefining resilience and capability in the Industry 5.0 era.